Stroke Pathway: acute management to secondary prevention

Adult ischaemic stroke, transient ischaemic attack and intracerebral haemorrhage, built from the Stroke Foundation *Living Clinical Guidelines for Stroke Management*, *Therapeutic Guidelines: Neurology* and the Australian Commission on Safety and Quality in Health Care *Acute Stroke Clinical Care Standard*. Subarachnoid haemorrhage is out of scope.

Ischaemic pathway · Haemorrhagic pathway · Decision point · Medication (dose · class · mechanism) · Secondary prevention / Australian pearl

STAGE 1 Recognition and emergency assessment

Pre-hospital recognition

- **F**ace drooping, **A**rm weakness, **S**peech difficulty, **T**ime to call triple zero (000) — the Stroke Foundation FAST campaign.
- Paramedics use a validated stroke screen and large vessel occlusion screen, record **time last known well**, check capillary blood glucose and pre-notify the receiving stroke centre (code stroke). Mobile stroke units operate in Melbourne and Adelaide.
- Telestroke networks (for example Victorian Stroke Telemedicine, NSW Telestroke Service) support rural sites.

`Door-to-needle target < 60 min`

Emergency department: code stroke

- Airway, breathing, circulation; oxygen only if saturation < 95%.
- Exclude hypoglycaemia (< 4 mmol/L) and other mimics (seizure with Todd paresis, migraine, functional, sepsis).
- National Institutes of Health Stroke Scale (NIHSS) score for severity.
- **Immediate imaging:** non-contrast computed tomography (CT) brain + CT angiography (aortic arch to vertex) ± CT perfusion; magnetic resonance imaging (MRI) where rapidly available.
- Bloods (do not delay thrombolysis waiting for them unless anticoagulant use or bleeding disorder suspected): full blood count, electrolytes and creatinine, coagulation profile, glucose, troponin; 12-lead electrocardiogram (ECG).
- Nil by mouth until swallow screen passed.

Have the symptoms fully resolved?

Yes → probable transient ischaemic attack: go to the transient ischaemic attack pathway. **No** (any persisting deficit) → treat as stroke and continue below.

`still symptomatic`

CT brain: is there haemorrhage?

CT cannot reliably distinguish infarct early; a normal CT in the first hours is expected in ischaemic stroke.

STAGE 1 CONT.

Ischaemic vs haemorrhagic management

NO HAEMORRHAGE → ISCHAEMIC STROKE

Eligible for reperfusion?

Potentially disabling deficit, onset or last known well within window, no contraindication.

yes

≤ 4.5 h from last known well

Intravenous thrombolysis

Tenecteplase 0.25 mg/kg IV bolus (max 25 mg)

Fibrinolytic (genetically modified tissue plasminogen activator): binds fibrin and converts plasminogen to plasmin, lysing clot. Greater fibrin specificity and longer half-life than alteplase allow a single bolus. Preferred in many Australian centres.

Or alteplase 0.9 mg/kg (max 90 mg): 10% bolus, remainder over 60 min

Fibrinolytic (recombinant tissue plasminogen activator): same mechanism, short half-life needs infusion.

- 4.5–9 h or wake-up stroke: thrombolysis may be considered when CT perfusion or MRI shows salvageable penumbra (EXTEND, WAKE-UP trials).
- Blood pressure must be < 185/110 mmHg before and < 180/105 mmHg for 24 h after.
- Key contraindications: any intracranial haemorrhage, recent major surgery / head trauma / prior intracranial bleed, active bleeding, platelets < 100 × 10^9/L, international normalised ratio (INR) > 1.7, therapeutic direct oral anticoagulant within 48 h (unless reversed or levels low per local protocol).
- Neuro observations and blood pressure every 15 min for 2 h, then 30-minutely; watch for angio-oedema (more likely with angiotensin converting enzyme inhibitors) and haemorrhage.

Do not delay thrombolysis while arranging thrombectomy.

≤ 6 h, or 6–24 h with imaging selection

Endovascular thrombectomy

- Large vessel occlusion of internal carotid artery or proximal middle cerebral artery (M1, selected M2): within 6 h; 6–24 h if perfusion imaging shows mismatch (DAWN, DEFUSE-3).
- Basilar artery occlusion: within 24 h (ATTENTION, BAOCHE).
- Large ischaemic core (low Alberta Stroke Program Early CT Score, ASPECTS 3–5): now supported by trials (SELECT2, ANGEL-ASPECT, RESCUE-Japan LIMIT).
- Medium and distal vessel occlusions: recent trials (ESCAPE-MeVO, DISTAL) showed no benefit, so not routine.
- Transfer to a comprehensive (thrombectomy-capable) stroke centre; "drip and ship" if needed.
- After successful reperfusion, do **not** intensively lower systolic below 140 mmHg (harm in ENCHANTED2/MT, OPTIMAL-BP).

HAEMORRHAGE → INTRACEREBRAL HAEMORRHAGE

Anticoagulated? Size and site?

CT angiography for spot sign / underlying lesion (aneurysm, arteriovenous malformation). Glasgow Coma Scale and ICH score for prognosis.

Within 1 h of arrival

Rapid blood pressure lowering

- If systolic 150–220 mmHg: target systolic **~140 mmHg** within 1 hour and keep it stable (INTERACT2, INTERACT3 care bundle). Avoid large, abrupt falls and systolic < 110 mmHg.

Labetalol 10–20 mg IV over 2 min, repeat every 10 min (max 300 mg) or infusion

Combined alpha-1 and non-selective beta-adrenergic blocker: reduces peripheral resistance without reflex tachycardia.

Glyceryl trinitrate IV infusion (titrated) or hydralazine 5–10 mg IV

Glyceryl trinitrate — organic nitrate: nitric oxide donor, raises cyclic guanosine monophosphate, mainly venodilation. **Hydralazine — direct arteriolar vasodilator**.

INTERACT3 bundle (blood pressure, glucose, temperature, anticoagulation reversal) reduced death and disability. These same agents are used to bring blood pressure under 185/110 before thrombolysis.

`Within 48 h (24 h after lysis, once haemorrhage excluded)`

Early antiplatelet therapy

Aspirin 300 mg stat (oral, or rectal if swallow unsafe), then 100 mg daily
Antiplatelet, salicylate: irreversibly acetylates cyclo-oxygenase-1, blocking platelet thromboxane A2 production for the platelet's life (7–10 days).

Minor stroke (NIHSS ≤ 3) or high-risk transient ischaemic attack, no thrombolysis and no cardioembolic source: dual antiplatelet therapy started within 24 h for **21 days**, then single agent (CHANCE, POINT).

Clopidogrel 300–600 mg load, then 75 mg daily + aspirin 100 mg
Antiplatelet, thienopyridine P2Y12 inhibitor: prodrug activated by hepatic cytochrome P450 2C19; irreversibly blocks the P2Y12 adenosine diphosphate receptor, preventing glycoprotein IIb/IIIa activation.

Alternative: ticagrelor 180 mg load, then 90 mg twice daily + aspirin (30 days, THALES; CHANCE-2)
Antiplatelet, cyclopentyl-triazolopyrimidine P2Y12 inhibitor: direct-acting, reversible P2Y12 blockade; useful if a cytochrome P450 2C19 loss-of-function carrier.

The same dual antiplatelet regimen applies to high-risk transient ischaemic attack; see the transient ischaemic attack pathway.

Ischaemic-specific acute care

- **Blood pressure** if not reperfused: lower only if > 220/120 mmHg, and by no more than about 15% in 24 h.
- **Malignant middle cerebral artery infarction**: neurosurgical referral for decompressive hemicraniectomy within 48 h (strongest benefit age ≤ 60).
- **Cerebellar infarct** with mass effect or hydrocephalus: neurosurgery (ventricular drain / suboccipital decompression).

`Immediately`

Reverse anticoagulation

AGENT	REVERSAL (AUSTRALIAN CONSENSUS)
Warfarin	**Prothrombinex-VF 50 units/kg IV** **Three-factor prothrombin complex concentrate** (factors II, IX, X): directly replaces vitamin K-dependent clotting factors. **+ Vitamin K (phytomenadione) 5–10 mg IV** **Fat-soluble vitamin**: cofactor for hepatic gamma-carboxylation of factors II, VII, IX, X; sustains reversal after the concentrate wears off. Fresh frozen plasma only if Prothrombinex-VF unavailable. Recheck INR at 30 min.
Dabigatran	**Idarucizumab 5 g IV** **Humanised monoclonal antibody fragment**: binds dabigatran with very high affinity, neutralising it within minutes.
Apixaban, rivaroxaban	**Prothrombin complex concentrate 50 units/kg per local protocol; andexanet alfa where available** **Andexanet alfa — recombinant modified factor Xa decoy**: catalytically inactive, binds and sequesters factor Xa inhibitors.
Unfractionated heparin	**Protamine sulfate 1 mg per 100 units heparin given in last 2–3 h (max 50 mg)** **Heparin antagonist**: strongly basic protein that binds acidic heparin, forming an inactive complex.

Tranexamic acid is not routinely recommended for spontaneous intracerebral haemorrhage (TICH-2). Platelet transfusion for patients on antiplatelets is harmful (PATCH).

Neurosurgical referral

- Cerebellar haemorrhage > 3 cm, brainstem compression or hydrocephalus → surgical evacuation / external ventricular drain.
- Selected superficial lobar haematomas (30–80 mL) within 24 h: minimally invasive evacuation may improve outcome (ENRICH).
- Intraventricular extension with hydrocephalus → external ventricular drain.
- Venous thromboembolism prophylaxis: intermittent pneumatic compression from admission; consider low molecular weight heparin after 48 h once haematoma stable.

STAGE 2

Stroke unit care (all stroke types)

Admit to a dedicated **stroke unit** with a multidisciplinary team; this alone reduces death and dependency. Follow the Australian **Quality in Acute Stroke Care (QASC)** "fever, sugar, swallow" protocols.

Fever

Paracetamol 1 g oral/IV 6-hourly for temperature ≥ 37.5 °C

Analgesic and antipyretic: central cyclo-oxygenase inhibition reduces hypothalamic prostaglandin E2.

Sugar

Insulin (subcutaneous or infusion) if glucose > 10 mmol/L

Hormone, glucose-lowering: promotes cellular glucose uptake via glucose transporter 4 and suppresses hepatic glucose output. Avoid hypoglycaemia.

- **Swallow:** screen before any oral food, fluid or medication; speech pathology assessment if failed. Nasogastric feeding if unsafe beyond a few days.
- **Venous thromboembolism prevention** (immobile patients): intermittent pneumatic compression (CLOTS 3). Graduated compression stockings are not recommended.

  Enoxaparin 40 mg subcutaneous daily (ischaemic stroke, high risk, weighed against bleeding)

  Low molecular weight heparin: potentiates antithrombin, mainly inhibiting factor Xa.
- **Mobilisation:** avoid very early, high-dose mobilisation within 24 h (AVERT); then short, frequent sessions.
- Screen for mood (depression, anxiety), cognition, aphasia, continence, pressure areas, falls, delirium and nutrition. Start rehabilitation early.
- Investigate stroke mechanism (below) and start secondary prevention **before discharge**; discharge with a care plan and enrol in the Australian Stroke Clinical Registry.

What caused the stroke?

Work-up: carotid duplex or CT angiography, ECG and inpatient cardiac monitoring, echocardiography (transthoracic ± transoesophageal with bubble study if young), lipids, glycated haemoglobin (HbA1c); consider thrombophilia / vasculitis screen in young patients. Classified using the TOAST (Trial of Org 10172 in Acute Stroke Treatment) scheme.

STAGE 2 CONT.

Secondary prevention by stroke mechanism

Atrial fibrillation / cardioembolic

Oral anticoagulation, **not** antiplatelets. Direct oral anticoagulants preferred over warfarin.

Apixaban 5 mg twice daily
(2.5 mg if 2 of: age ≥ 80, weight ≤ 60 kg, creatinine ≥ 133 µmol/L)
Direct factor Xa inhibitor.

Rivaroxaban 20 mg daily with food
(15 mg if creatinine clearance 15–49 mL/min)
Direct factor Xa inhibitor.

Dabigatran 150 mg twice daily
(110 mg if age ≥ 75 / bleeding risk; avoid if clearance < 30)
Direct thrombin (factor IIa) inhibitor.

Warfarin (INR 2–3)
for mechanical valves or moderate–severe mitral stenosis
Vitamin K antagonist: inhibits vitamin K epoxide reductase, reducing factors II, VII, IX, X.

Timing: early start is supported by recent trials (ELAN, OPTIMAS, TIMING): roughly within 48 h for minor–moderate stroke and day 6–7 for major stroke. Check current guideline wording and local protocol.

Cryptogenic stroke: prolonged rhythm monitoring (for example implantable loop recorder) to detect paroxysmal atrial fibrillation.

Anticoagulation contraindicated: left atrial appendage occlusion.

Large artery (carotid) atherosclerosis

Symptomatic internal carotid stenosis, graded by North American Symptomatic Carotid Endarterectomy Trial (NASCET) criteria:

- **70–99%:** carotid endarterectomy, ideally **within 2 weeks** of the event, at a low-complication centre.
- **50–69%:** consider endarterectomy in selected patients (greater benefit in men, older patients, recent event).
- **< 50% or occlusion:** no surgery; best medical therapy.
- Carotid stenting: alternative for selected patients (restenosis, radiation-induced stenosis, surgically hostile neck); higher periprocedural stroke risk in older patients.

Plus long-term antiplatelet, high-intensity statin and blood pressure control.

Asymptomatic stenosis: best medical therapy is the default.

Small vessel (lacunar) / other non-cardioembolic

Long-term single antiplatelet therapy (one of):

Aspirin 100 mg daily
Cyclo-oxygenase-1 inhibitor (salicylate antiplatelet).

Clopidogrel 75 mg daily
P2Y12 inhibitor (thienopyridine antiplatelet).

Aspirin 25 mg + modified-release dipyridamole 200 mg twice daily
Dipyridamole — phosphodiesterase inhibitor and adenosine reuptake inhibitor: raises platelet cyclic adenosine monophosphate. Headache common.

Long-term aspirin + clopidogrel beyond the first weeks increases bleeding without benefit (MATCH, CHARISMA).

Rare causes (dissection, vasculitis, antiphospholipid syndrome) need cause-specific treatment.

Embolic stroke of undetermined source + patent foramen ovale

- Age 18–60 with no other cause and a patent foramen ovale with high-risk features (atrial septal aneurysm or large shunt): **percutaneous closure + antiplatelet** (CLOSE, RESPECT, REDUCE).
- Embolic stroke of undetermined source without atrial fibrillation: antiplatelet; empirical anticoagulation not superior (NAVIGATE ESUS, RE-SPECT ESUS).

After intracerebral haemorrhage

- **Blood pressure control is the most important measure**; target < 130/80 mmHg.
- Restarting an antiplatelet for a strong vascular indication appears safe (RESTART).
- Atrial fibrillation after haemorrhage: individualise with the neurologist (consider haemorrhage location, cerebral amyloid angiopathy markers); options include delayed direct oral anticoagulant or left atrial appendage occlusion.
- Limit alcohol; avoid sympathomimetic drugs.

STAGE 3

Secondary prevention for every patient

Blood pressure

Start or optimise before discharge once stable. Target **< 130/80 mmHg** for most patients (lower targets reduce recurrence; PROGRESS showed benefit even if normotensive).

Perindopril 4–8 mg daily
Angiotensin converting enzyme inhibitor: blocks conversion of angiotensin I to II, reducing vasoconstriction and aldosterone.

+ Indapamide 1.25–2.5 mg daily
Thiazide-like diuretic: inhibits sodium-chloride cotransporter in distal convoluted tubule.

Or candesartan / amlodipine
Candesartan — angiotensin II receptor blocker (type 1 receptor). **Amlodipine — dihydropyridine calcium channel blocker**: blocks L-type channels in arteriolar smooth muscle.

Lipids (ischaemic stroke / transient ischaemic attack)

Target low-density lipoprotein (LDL) cholesterol **< 1.8 mmol/L** (Treat Stroke to Target trial).

Atorvastatin 40–80 mg daily (or rosuvastatin 20–40 mg)
Statin, HMG-CoA (3-hydroxy-3-methylglutaryl coenzyme A) reductase inhibitor: cuts hepatic cholesterol synthesis, upregulating LDL receptors; plaque-stabilising.

+ Ezetimibe 10 mg daily if above target
Cholesterol absorption inhibitor: blocks Niemann-Pick C1-Like 1 transporter in the small intestine.

Evolocumab (Pharmaceutical Benefits Scheme criteria)
PCSK9 (proprotein convertase subtilisin/kexin type 9) inhibitor monoclonal antibody: stops LDL receptor degradation.

Diabetes

Individualised HbA1c, usually ≤ 7% (53 mmol/mol). Prefer agents with proven cardiovascular benefit.

Metformin
Biguanide: activates AMP-activated protein kinase, reducing hepatic gluconeogenesis.

Glucagon-like peptide-1 receptor agonist (semaglutide, dulaglutide)
Incretin mimetic: glucose-dependent insulin release, less glucagon, slower gastric emptying, weight loss; reduced stroke in cardiovascular outcome trials.

Sodium-glucose cotransporter 2 inhibitor (empagliflozin)
Gliflozin: blocks proximal tubule glucose reabsorption; heart failure and kidney benefit.

Smoking

Brief advice + Quitline (13 78 48) + pharmacotherapy.

Varenicline 1 mg twice daily (after titration), 12 weeks
Partial agonist at α4β2 nicotinic acetylcholine receptors: eases withdrawal and blunts reward from smoking.

Nicotine replacement therapy (patch + short-acting form)
Nicotinic receptor agonist: replaces nicotine without tobacco toxins.

Lifestyle

- Alcohol: ≤ 10 standard drinks a week and ≤ 4 on any day (National Health and Medical Research Council).
- Physical activity: 150–300 min moderate activity a week as able; reduce sitting.
- Mediterranean-style diet, salt < 5 g/day, weight management.
- Screen for obstructive sleep apnoea.
- Avoid oestrogen-containing contraception / hormone therapy where possible.

Driving, follow-up and return to life

- **Austroads *Assessing Fitness to Drive*:** private licence, no driving for at least 4 weeks after stroke and 2 weeks after transient ischaemic attack; commercial licence at least 3 months after stroke. Longer if visual field loss, cognitive or motor deficit.
- Review by general practitioner within weeks of discharge; GP management plan and team care arrangements for allied health.
- Ongoing rehabilitation, return to work, carer support, StrokeLine (1800 787 653).
- Check adherence at every visit; it is the biggest cause of failed secondary prevention.

SIDE PATHWAY

Transient ischaemic attack (TIA)

What it is and why it is urgent

- **Tissue-based definition:** a transient episode of neurological dysfunction caused by focal brain, spinal cord or retinal ischaemia **without acute infarction**. Most episodes last < 1 hour. If magnetic resonance imaging (MRI) with diffusion-weighted imaging shows an infarct, it is a stroke, even if symptoms resolved.
- Stroke risk is front-loaded: roughly half of the 90-day risk falls in the first 48 hours. Urgent assessment and treatment cut 90-day stroke risk by about 80% (EXPRESS, SOS-TIA studies).
- Retinal TIA (amaurosis fugax: painless, transient monocular vision loss, "curtain coming down") is managed the same way.

Common mimics: migraine aura (positive, spreading symptoms), focal seizure, syncope, hypoglycaemia, transient global amnesia, peripheral vertigo, functional symptoms. Isolated dizziness, confusion or loss of consciousness are rarely a TIA.

Symptoms still present, or recurring right now?

Yes → this is a stroke until proven otherwise: call triple zero (000) / code stroke and go back to **Stage 1** (may be a thrombolysis candidate). **No** → urgent TIA assessment below.

fully resolved

`Specialist assessment within 24 h`

Risk stratify and decide where to assess

The Stroke Foundation recommends that **all** patients with suspected TIA get urgent assessment and imaging, ideally within 24 hours, through an emergency department or rapid-access TIA clinic. Admit or seek immediate specialist review if any high-risk feature is present:

- Crescendo TIA (2 or more episodes within a week)
- Known or newly found atrial fibrillation, or already on an anticoagulant
- Symptomatic carotid stenosis already known
- Motor weakness or speech disturbance, or symptoms lasting > 10 minutes
- Investigations cannot be done promptly as an outpatient

ABCD² SCORE (AGE, BLOOD PRESSURE, CLINICAL FEATURES, DURATION, DIABETES)	
Age ≥ 60 years	1
Blood pressure ≥ 140/90 mmHg at first assessment	1
Clinical: unilateral weakness	2
Clinical: speech disturbance without weakness	1
Duration ≥ 60 min / 10–59 min	2 / 1
Diabetes	1

Score ≥ 4 is "high risk", but do not use ABCD² alone to decide who gets urgent care: it misses atrial fibrillation and carotid stenosis, which drive the highest early risk.

`Complete within 24 h`

Investigations

- **Brain imaging:** MRI with diffusion-weighted imaging preferred (detects infarct in up to about 1 in 3, which predicts recurrence); non-contrast computed tomography (CT) if MRI not available, mainly to exclude haemorrhage and mimics.
- **Vessel imaging:** CT angiography (arch to vertex) or carotid duplex ultrasound for carotid and intracranial stenosis.
- **Cardiac:** 12-lead electrocardiogram (ECG); Holter or prolonged rhythm monitoring if no cause found; echocardiography if cardioembolism suspected or patient is young.
- **Bloods:** full blood count, electrolytes and creatinine, glucose and glycated haemoglobin (HbA1c), fasting lipids, coagulation profile.
- Amaurosis fugax in a patient ≥ 50: check erythrocyte sedimentation rate and C-reactive protein for **giant cell arteritis**, and arrange urgent ophthalmology review.

`Start immediately once haemorrhage excluded`

Immediate treatment

High-risk TIA (ABCD² ≥ 4) or minor stroke, non-cardioembolic: dual antiplatelet therapy within 24 h of onset for **21 days**, then single antiplatelet (CHANCE, POINT).

Aspirin 300 mg stat, then 100 mg daily
Antiplatelet, salicylate: irreversibly inhibits cyclo-oxygenase-1, blocking platelet thromboxane A2 production.

+ Clopidogrel 300–600 mg load, then 75 mg daily
Antiplatelet, thienopyridine P2Y12 inhibitor: prodrug (activated by cytochrome P450 2C19) that irreversibly blocks the platelet adenosine diphosphate P2Y12 receptor.

Lower-risk TIA: single antiplatelet (aspirin 100 mg, clopidogrel 75 mg, or aspirin + modified-release dipyridamole) long term.

TIA with atrial fibrillation: start an oral anticoagulant promptly (no infarct, so bleeding risk is low; usually day 1). Do not add an antiplatelet.

Apixaban 5 mg twice daily (or rivaroxaban / dabigatran)
Direct oral anticoagulant: apixaban and rivaroxaban directly inhibit factor Xa; dabigatran directly inhibits thrombin.

Start at the same visit: high-intensity statin and blood pressure lowering (see **Stage 3**).

Atorvastatin 40–80 mg daily
Statin, HMG-CoA (3-hydroxy-3-methylglutaryl coenzyme A) reductase inhibitor: lowers low-density lipoprotein cholesterol and stabilises plaque.

Cause identified?

Manage exactly as for ischaemic stroke: secondary prevention by mechanism. Symptomatic carotid stenosis 70–99% (North American Symptomatic Carotid Endarterectomy Trial criteria) → carotid endarterectomy within 2 weeks, ideally sooner; risk of recurrence is highest in the first days.

Before the patient leaves

- **Safety-net:** if symptoms return, call triple zero (000) immediately; do not wait to see if they settle. Teach FAST (Face, Arms, Speech, Time).
- **Driving (Austroads *Assessing Fitness to Drive*):** private licence, no driving for at least 2 weeks after a TIA; commercial licence, longer stand-down (check current standard). Recurrent TIAs need specialist clearance.
- Written plan of medications and follow-up; letter to the general practitioner; neurology or TIA clinic follow-up.
- Lifestyle measures and smoking cessation as in Stage 3.

▼ Abbreviations and trial names

ASPECTS Alberta Stroke Program Early CT Score

CT computed tomography

ECG electrocardiogram

HbA1c glycated haemoglobin

INR international normalised ratio

LDL low-density lipoprotein

MRI magnetic resonance imaging

NASCET North American Symptomatic Carotid Endarterectomy Trial

NIHSS National Institutes of Health Stroke Scale

PCSK9 proprotein convertase subtilisin/kexin type 9

QASC Quality in Acute Stroke Care

TOAST Trial of Org 10172 in Acute Stroke Treatment

AVERT A Very Early Rehabilitation Trial

CHANCE Clopidogrel in High-risk patients with Acute Non-disabling Cerebrovascular Events

POINT Platelet-Oriented Inhibition in New TIA and minor ischaemic stroke

THALES Ticagrelor and Aspirin or Aspirin Alone in Acute Ischemic Stroke or TIA

INTERACT Intensive Blood Pressure Reduction in Acute Cerebral Haemorrhage Trial

ENRICH Early Minimally Invasive Removal of Intracerebral Hemorrhage

ELAN Early versus Late initiation of direct oral Anticoagulants in post-ischaemic stroke patients with atrial fibrillation

OPTIMAS Optimal Timing of Anticoagulation After Acute Ischaemic Stroke

ESUS embolic stroke of undetermined source

TIA transient ischaemic attack

ABCD² Age, Blood pressure, Clinical features, Duration, Diabetes

EXPRESS Early use of Existing Preventive Strategies for Stroke

SOS-TIA a Paris 24-hour-access TIA clinic study

PROGRESS Perindopril Protection Against Recurrent Stroke Study

▼ Sources

Stroke Foundation. *Living Clinical Guidelines for Stroke Management* (Australia and New Zealand), via InformMe.

Therapeutic Guidelines Limited. *Therapeutic Guidelines: Neurology* (eTG complete), stroke and transient ischaemic attack topics.

Australian Commission on Safety and Quality in Health Care. *Acute Stroke Clinical Care Standard* (2019, updated).

Bladin CF, et al. Australian and New Zealand living clinical guidelines for stroke management: update. Med J Aust 2022;216(9).

Tran HA, et al. An update of consensus guidelines for warfarin reversal. Med J Aust 2013;198(4):198–199 (Australasian Society of Thrombosis and Haemostasis).

Austroads. *Assessing Fitness to Drive* (2022).

Middleton S, et al. QASC trial. Lancet 2011;378:1699–1706. Werring DJ, et al. OPTIMAS. Lancet 2024. Goyal M, et al. ESCAPE-MeVO. N Engl J Med 2025. Large-core thrombectomy review: PMC12418464.

> Study aid for medical students. The Stroke Foundation guideline is "living" and updated as trials report; confirm doses and eligibility against the current guideline, eTG and your hospital's stroke protocol before clinical use.